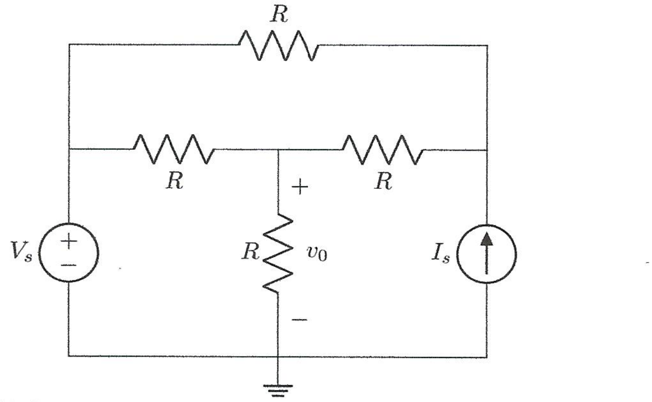
(*) EXAM MEMO

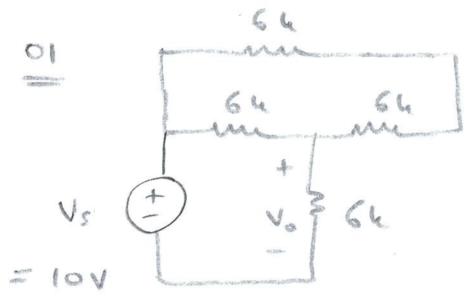
Section 1

Questions 1 to 2 [4]



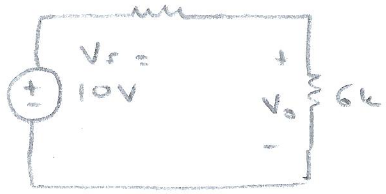
01	Determine the contribution to v_0 of the voltage source V_s alone if $I_s = 1$ mA and $R = 6$ kΩ. [2] (The value of V_s is given in ClickUP.)
02	Determine the contribution to v_0 of the current source I_s alone if $V_s = 10$ V. [2] (The values of I_s and R are given in ClickUP.)

01



$\Rightarrow$

$(6+6)\|6 = 4\,k\Omega$

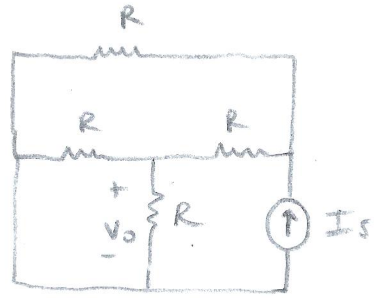


$V_s = 10V$

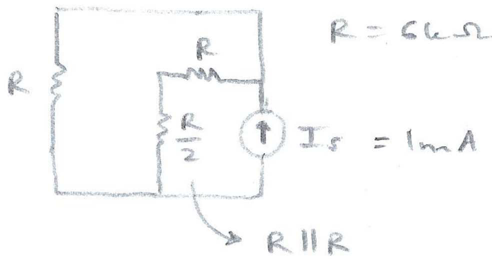
$$\therefore V_o = V_s\left(\frac{6k}{6k+4k}\right) = \frac{3}{5}V_s\ [V]\ (*)$$

$$= \frac{3}{5}(10) = \underline{6V}$$

02



$\Rightarrow$



$R = 6k\Omega$

$I_s = 1mA$

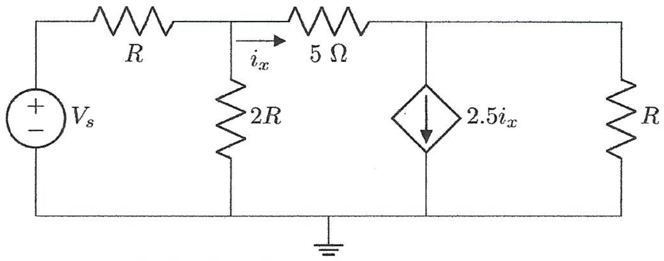
$R = 6k\Omega$

$$i_o = I_s\left(\frac{R}{R + \left(\frac{R}{2} + R\right)}\right) = \frac{I_s R}{\frac{5}{2}R} = \frac{2}{5}I_s$$

$$V_o = \frac{2}{5}I_s \times \frac{R}{2} = \frac{I_s R}{5}\ [V]\ (*) = \frac{(1\,mA)(6\,k\Omega)}{5}$$

$$= \underline{1.2V}$$

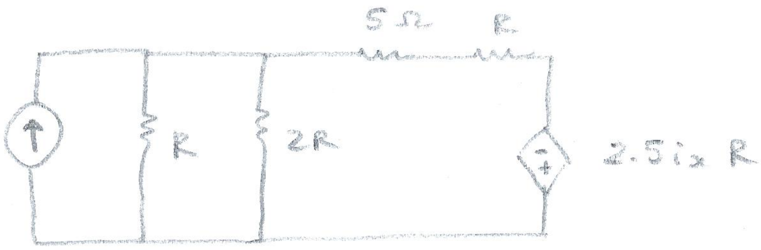
Question 3 [2]



03	Use source transformation to determine i_x. [2] (The values of V_s and R are given in ClickUP.)

$V_s = 18\text{ V}, \quad R = 12\ \Omega:$

$\frac{V_s}{R}$



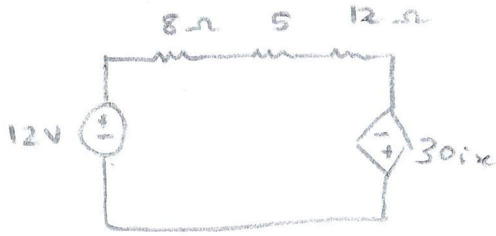
$I = \frac{18}{12} = 1.5\text{A}$

$12 \parallel 24 = 8\ \Omega$

$\Rightarrow$

$\frac{V_s}{R} \times \frac{2R}{3}$

$= \frac{2}{3} V_s$



$$i_x = \frac{\frac{2}{3}V_s + 2.5 i_x R}{\frac{2}{3}R + 5 + R}$$

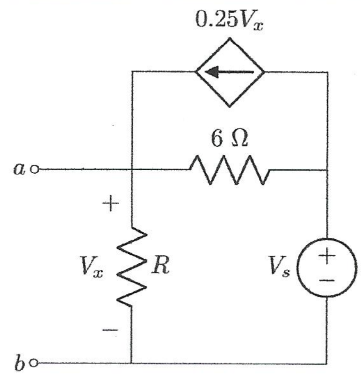
$$i_x = \frac{12 + 30 i_x}{25}$$

$$25 i_x = 12 + 30 i_x$$

$$i_x = -\frac{12}{5}$$

$$i_x = \frac{\frac{2}{3}V_s}{\left(5 - \frac{5}{6}R\right)} \quad [A] \quad *$$

$$\therefore i_x = -2.4\text{ A}$$

Questions 4 to 5 [4]	
Find the Thevenin equivalent voltage V_{TH} and Thevenin equivalent resistance R_{TH}, with reference to terminals $a-b$. (The values of V_s and R are given in ClickUP.)	
	
04	Determine V_{TH}. [2]
05	Determine R_{TH}. [2]

04

$V_{TH} = V_x$

$$\frac{V_x}{R} + \frac{V_x - V_s}{6} = 0.25\,V_x$$

$$V_{TH} = \frac{V_s R}{6 - 0.5R} \quad [V] \; *$$

$V_s = 45\text{ V}$

$R = 4\,\Omega$:

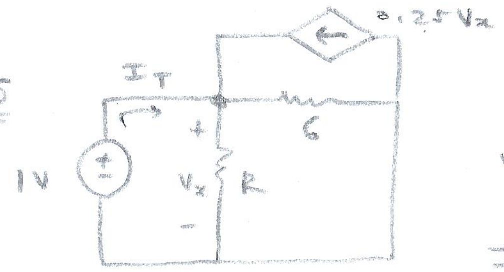
$$\frac{V_x}{4} + \frac{V_x - 45}{6} = 0.25\,V_x$$

$(\times 12)$: $3V_x + 2V_x - 90 = 3V_x$

$2V_x = 90$

$V_x = 45\text{ V}$

05



$V_x = 1\text{V}$

$$I_T = \frac{1}{R} + \frac{1}{6} - 0.25(1)$$

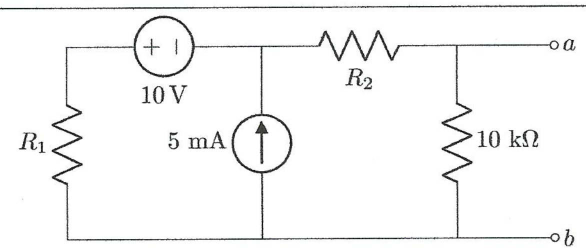
$$R_{TH} = \frac{1\text{V}}{I_T} = \frac{1}{\left(\frac{1}{R} + \frac{1}{6} - 0.25\right)}$$

$$\times 6R: \quad = \frac{6R}{6 - 0.5R} \quad [\Omega] \; *$$

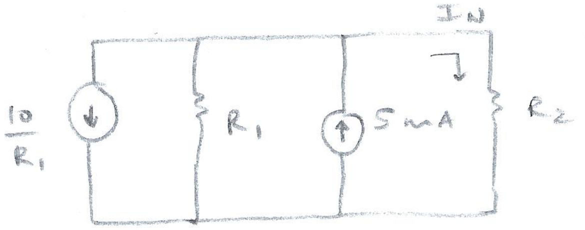
$R = 3$:

$$R_{TH} = \frac{1}{\frac{1}{3} + \frac{1}{6} - 0.25}$$

$R_{TH} = 4\,\Omega$

Questions 6 to 7 [4]	
Find the Norton equivalent current I_N and Norton equivalent resistance R_N, with reference to terminals $a-b$. (The values of R_1 and R_2 are given in ClickUP.)	
	
06	Determine I_N. [2]
07	Determine R_N. [2]

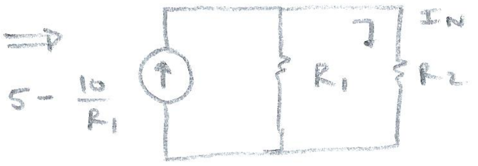
06 I_N:



$R_1 = 7k\Omega$

$R_2 = 3k\Omega$

$\Rightarrow$

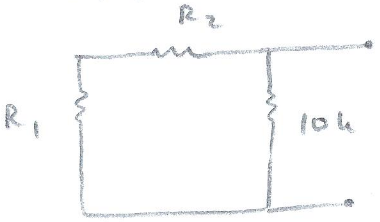


$$I_N = \left(5 - \frac{10}{R_1}\right)\left(\frac{R_1}{R_1 + R_2}\right)$$

$$I_N = \frac{5R_1 - 10}{R_1 + R_2} \quad [mA]$$

$$I_N = 3.572\left(\frac{7k}{10k}\right) = 2.5mA$$

07: R_N:



$R_1 = 7k$

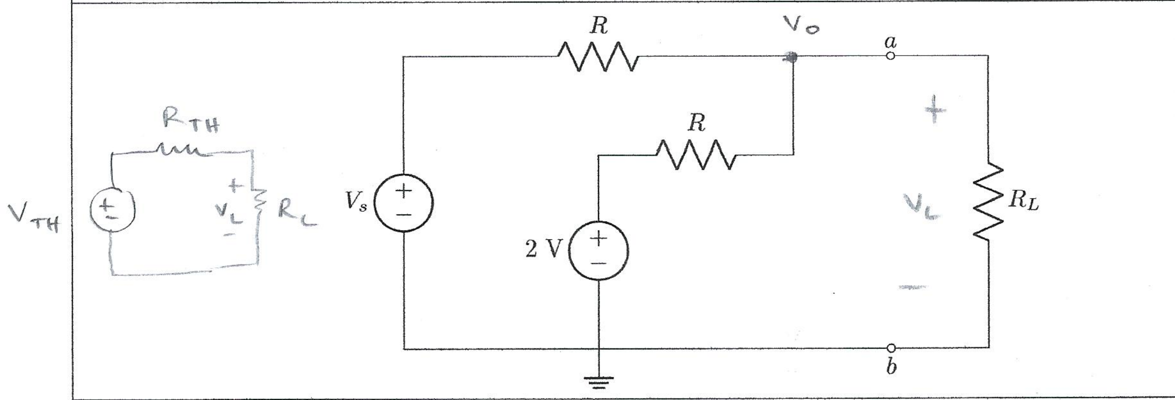
$R_2 = 3k$

$$R_N = (R_1 + R_2) \| 10k$$

$$R_N = \frac{10(R_1 + R_2)}{(10 + R_1 + R_2)} \quad [k\Omega]$$

$$R_N = \frac{10(10)}{20} = 5k\Omega$$

Questions 8 to 9 [4]



08	Determine the voltage V_L across the load R_L for maximum power transfer if $R = 250\ \Omega$. [2] (The value of V_s is given in ClickUP.)
09	Determine the value of R such that the maximum power P_{max} is delivered to the load, given that $V_s = V_L = 2$ V. [2] $V_s = 2V, \quad V_L = 1V$ (The value of P_{max} is given in ClickUP.)

08:

$$\frac{V_o - V_s}{R} + \frac{V_o - 2}{R} = 0$$

$$\therefore 2V_o = V_s + 2$$

$$V_o = \frac{V_s + 2}{2} = V_{TH}$$

$$V_L = \frac{V_{TH}}{2} \quad \text{(max power } R_{TH} = R_L\text{)} \qquad \therefore V_L = \frac{V_s + 2}{4}\ [V]$$

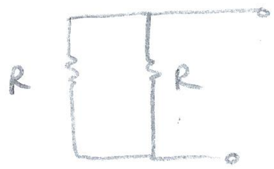
$V_s = 4V$:

$$\therefore V_L = 1.5\ V$$

09:

$$P_{max} = \frac{V_{TH}^2}{4R_{TH}} = \frac{(2V_L)^2}{4\left(\frac{R}{2}\right)} = \frac{\cancel{4} V_L^2}{\cancel{4}\frac{R}{2}}$$

$= 1 \quad (V_L = 1V)$



$$R_{TH} = \frac{R}{2}$$

$$P_{max} = \frac{4}{4\left(\frac{R}{2}\right)} = \frac{2}{R}$$

$$\therefore R = \frac{2}{P_{max}}\ [\Omega]$$

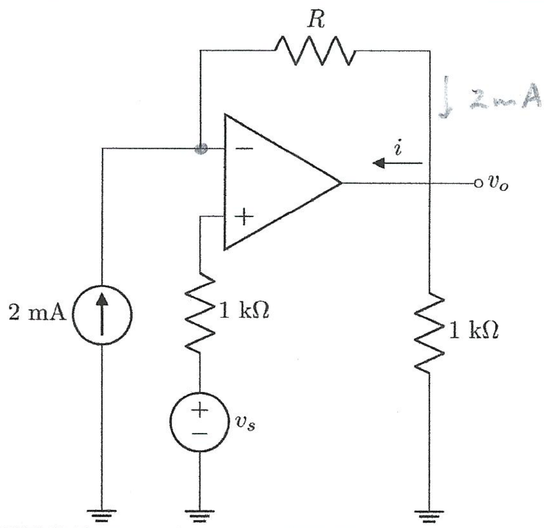
$P_{max} = 10\ mW$

$$\therefore P_{max} = \frac{2}{R}$$

$$R = \frac{2}{P_{max}}$$

$$R = \frac{2}{10\ mW} = 200\ \Omega$$

Question 10 [2]



10	Find i. [2] (The values of v_s and R are given in ClickUP.)

$V_s = 6V$

$R = 10k\Omega$

$$-2mA + \frac{V_s - V_o}{R} = 0$$

$$-2 + \frac{6 - V_o}{10} = 0$$

$$V_o = V_s - 2mA(R)$$

$$-20 + 6 = V_o$$

$$V_o = -14$$

$$i = 2mA - \frac{V_o}{1k\Omega}$$

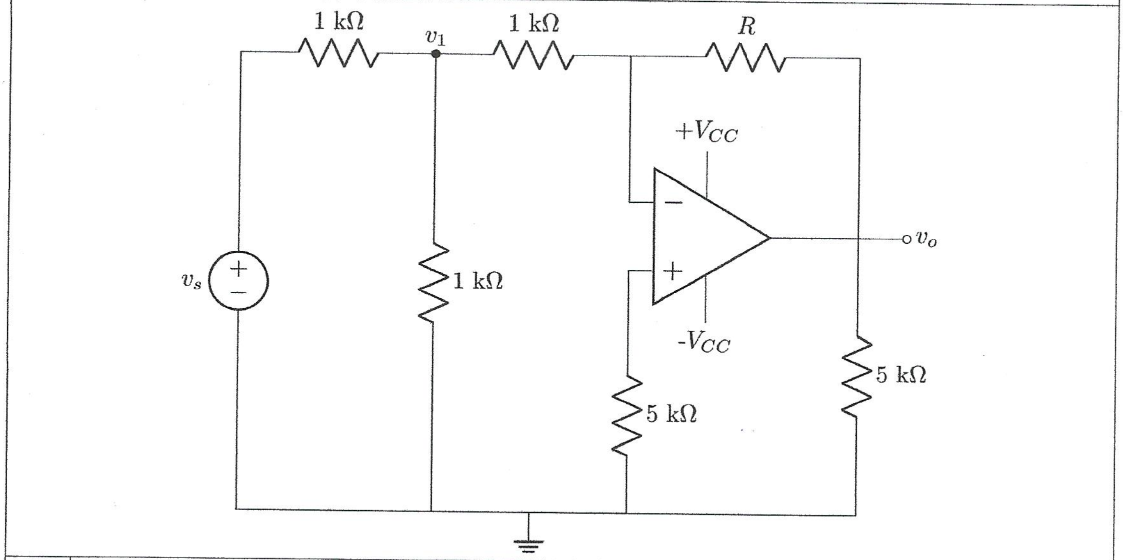
$$i = 2 - \left(\frac{-14}{1k}\right)$$

$$i = 2mA - \frac{V_s - 2mA(R)}{1k\Omega}$$

$$i = 16 mA$$

$$i = 2 - V_s + 2R \quad [mA] \quad (*)$$

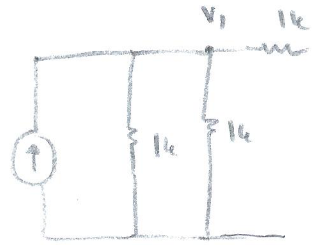
Questions 11 to 12 [4]



11	Determine v_o. [2] (The values of v_s and R are given in ClickUP.)
12	Determine the maximum value of v_1 before the op-amp goes into saturation. [2] (The values of V_{CC} and R are given in ClickUP.)

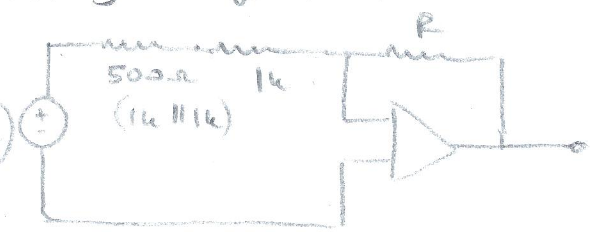
11

$\frac{V_s}{1k}$



inverting configuration:

$\frac{V_s}{1k}(500\Omega) = \frac{V_s}{2}$



$$V_o = -\frac{V_s}{2}\left(\frac{R}{1.5}\right)$$

$R = 10k\Omega$

$V_s = 3$

$$V_o = -\frac{V_s R}{3} \quad [V]$$

$$V_o = -\frac{3}{2}\left(\frac{10}{1.5}\right) = -10\text{ V}$$

12

$$-V_{cc} \leq V_o \leq V_{cc}$$

$R = 2k\Omega$

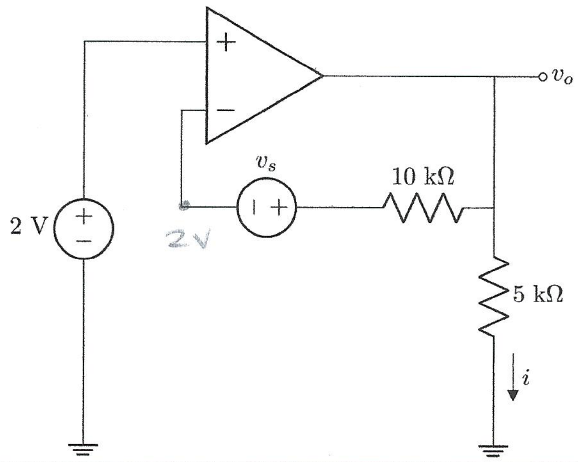
$V_{cc} = 12\text{ V}$

$$-V_{cc} \leq -\frac{R}{1k}V_1 \leq V_{cc}$$

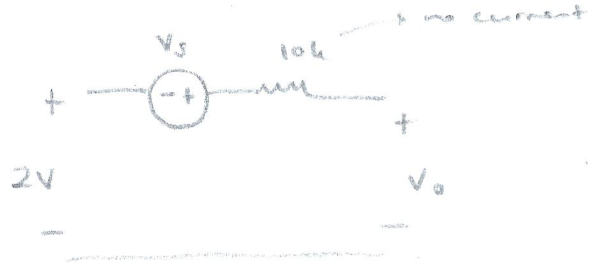
$$V_1 = \frac{V_{cc}}{R} \quad [V] \text{ max}$$

$$V_1 = \frac{12}{2} = 6\text{ V}$$

Question 13 [2]



13	Find i. [2] (The value of v_s is given in ClickUP.)



$V_s = 3\,V$

$V_o = 2 + 3 = 5$

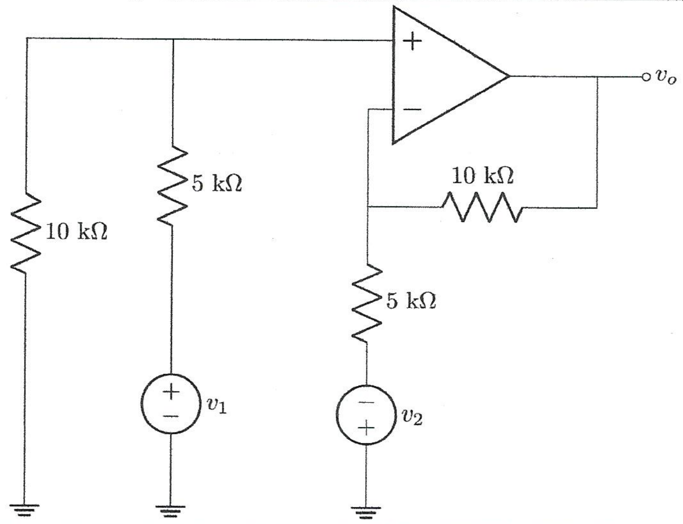
$i = \frac{5}{5} = 1\,mA$

$-2\,V - V_s + V_o = 0$

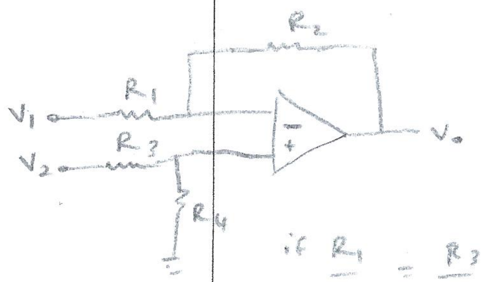
$V_o = 2 + V_s$

$i = \frac{V_o}{5k} = \frac{2 + V_s}{5}$ [mA] (*)

Question 14 [2]



14	Find v_o. [2] (The values of v_1 and v_2 are given in ClickUP.)



if $\frac{R_1}{R_2} = \frac{R_3}{R_4}$

$V_o = \frac{R_2}{R_1}(V_2 - V_1)$

$\Rightarrow V_o = 2(V_1 - (-V_2))$

difference amplifier:

$$V_o = \frac{R_2}{R_1}\left(V_1 - (-V_2)\right) \quad \text{as} \quad \frac{R_1}{R_2} = \frac{R_3}{R_4}$$

$$V_o = \frac{10k}{5k}(V_1 + V_2)$$

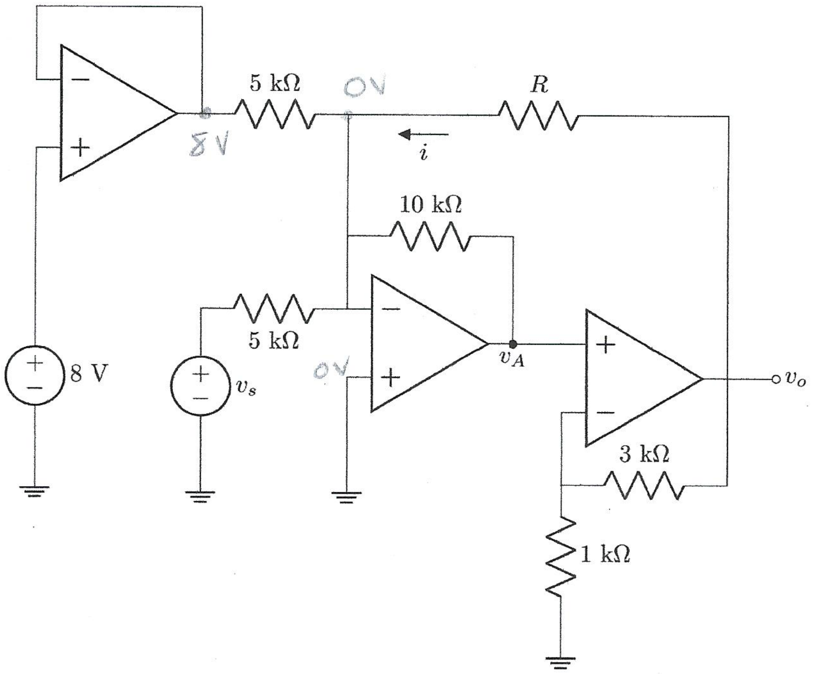
$$V_o = 2(V_1 + V_2)\ [V] \ \circledast$$

if $V_1 = 3\,V$

$V_2 = 2\,V$

$V_o = 10\,V$

Questions 15 to 16 [4]



15	Determine v_A when $R \to \infty$. [2] (The value of v_s is given in ClickUP.)
16	Determine i when $v_s = 6$ V. [2] (The values of R and v_A are given in ClickUP.)

15 summing amp: $V_A = -\left(\frac{10k}{5k} V_s + \frac{10k}{5k}(8)\right)$ $V_s = 6V$

$V_A = -(2V_s + 16)$

$V_A = -2(V_s + 8)$ [V] (*) $V_A = -(2(6) + 16)$

$V_A = -28$ V

16 non-inverting: $V_o = \left(1 + \frac{3k}{1k}\right) V_A$

and $i = \frac{V_o - 0}{R}$ $\therefore i = \frac{(1+3)V_A}{R} = \frac{4V_A}{R}$ [mA] (*)

Total Section 1 : [32] $V_A = 15$ V

Grand Total: [32] $R = 2k\Omega$

$V_o = 4(15)$

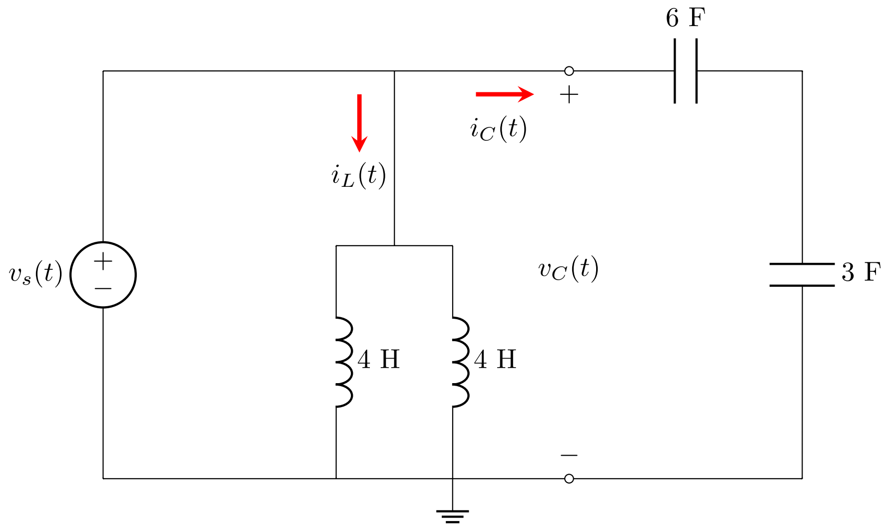
$= 60$

$i = \frac{60}{2} = 30$ mA

Section 3 / Afdeling 3

Questions 33 to 35 [6]

In the circuit below, $v_s(t) = \alpha e^{-0.2t}$ V and $i_L(0) = 2$ A.
The value of α is given in ClickUP.



33	Determine $v_C(0)$. [2]
34	Determine $i_C(5)$. [2]
35	Determine $i_L(5)$. [2]

33. $v_s(0) = v_c(0) \Rightarrow v_c(0) = \alpha$ (V).

34. $i_c(t) = C\frac{dv}{dt}$, $C = 6F \| 3F = 2F$. $v_c(t) = v_s(t)$

$\Rightarrow i_c(t) = 2\cdot[\alpha e^{-0.2t}]' = 2\alpha\cdot(-0.2)e^{-0.2t}$
$= -0.4e^{-0.2t}$ A

$\Rightarrow i_c(5) = -0.4\alpha e^{-1}$ A.

35. $i_L(t) = \frac{1}{L}\int_0^5 \alpha e^{-0.2t}\,dt + i_L(0)$. $L = 4\|4 = 2H$

$= \frac{\alpha}{2}\cdot\frac{1}{-0.2}\int_0^5 e^{-0.2t}\,d(-0.2t) + 2$

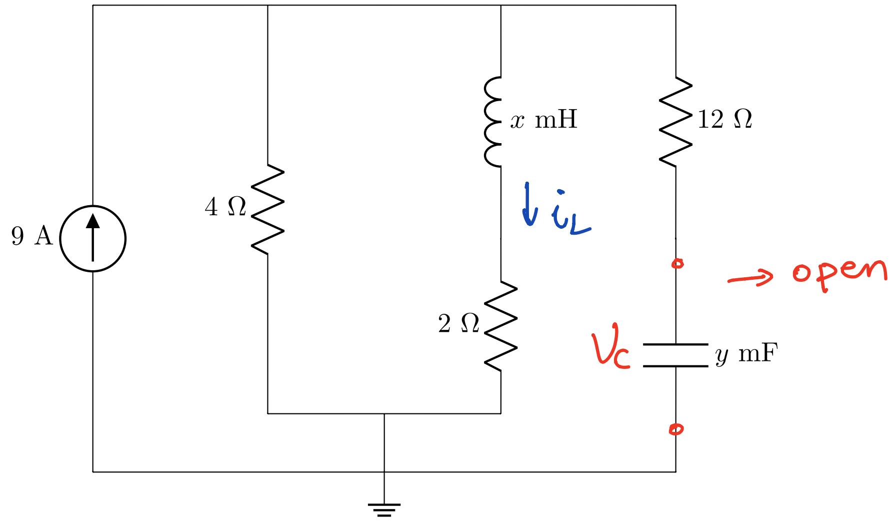
$= -2.5\alpha\left[e^{-0.2t}\right]_0^5 + 2$

$= -2.5\alpha(e^{-1} - 1) + 2$

$= 2.5\alpha - 2.5\alpha e^{-1} + 2$

Questions 36 to 37 [4]

The circuit below operates under steady-state DC conditions.
The values x and y are given in ClickUP.



36	Determine the energy stored in the inductor w_L. [2]
37	Determine the energy stored in the capacitor w_C. [2]

$$i_L = 9 \cdot \frac{4}{6} = 6A$$

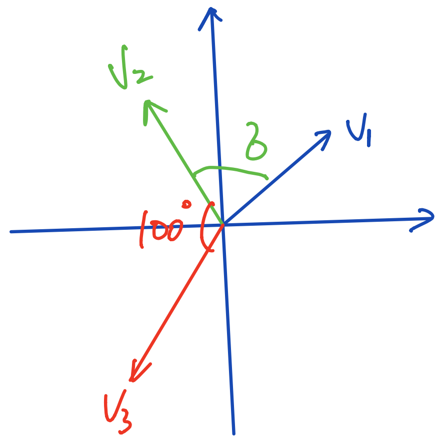
$$V_C = 2 \times 6 = 12V$$

36 $\Rightarrow W_L = \frac{1}{2} L i^2 = \frac{1}{2} x \cdot 10^{-3} \times 6^2 = 0.018x \, (J)$

37 $W_C = \frac{1}{2} C V^2 = \frac{1}{2} y \cdot 10^{-3} \cdot 12^2$

$$= 0.072y \, (J).$$

Question 38 [2]	
Sinusoids $v_1(t)$ and $v_2(t)$ have the following general form: $A\cos(\omega t + \phi)$. It is known that: - v_2 leads v_1 by σ°. - v_2 lags v_3 by 100°. The value of σ is given in ClickUP.	
38	What is the phase angle θ between v_1 and v_3 such that $0 \le \theta \le 180^\circ$. [2]



$\theta + \sigma + 100 = 360^\circ$

$\Rightarrow \theta = 260^\circ - \sigma.$

38

$0 \le 260^\circ - \sigma \le 180^\circ$

$\Rightarrow 80^\circ \le \sigma \le 260^\circ.$

Question 39 [2]	
A current source of $i(t) = 2\cos(100t + 90°)$ A is connected in parallel with a resistor of R Ω, a capacitor of C μF, and an inductor of L mH. Determine the phasor voltage $\mathbf{V}$ across the current source, assuming the sign of the voltage is positve. Values of R, C, and L are given in ClickUP.	
39	(a) If $\mathbf{V} = A + jB$, what is the real part A? [2]
39	(b) If $\mathbf{V} = A + jB$, what is the imaginary part B? [2]

$I = 2\angle 90° = j2.$ $\omega = 100.$

$Z_R = R$

$Z_L = j\omega L = j \cdot 100 \cdot L \times 10^{-3} = j0.1L$

$Z_C = -j\frac{1}{\omega C} = -j\frac{1}{100 \times C \times 10^{-6}} = -j\frac{10^4}{C}$

$Z_1 = Z_L \| Z_C = \dfrac{j0.1L \cdot -j\frac{10^4}{C}}{j0.1L - j\frac{10^4}{C}} = \dfrac{\frac{10^3 L}{C}}{j(0.1L - \frac{10^4}{C})}$

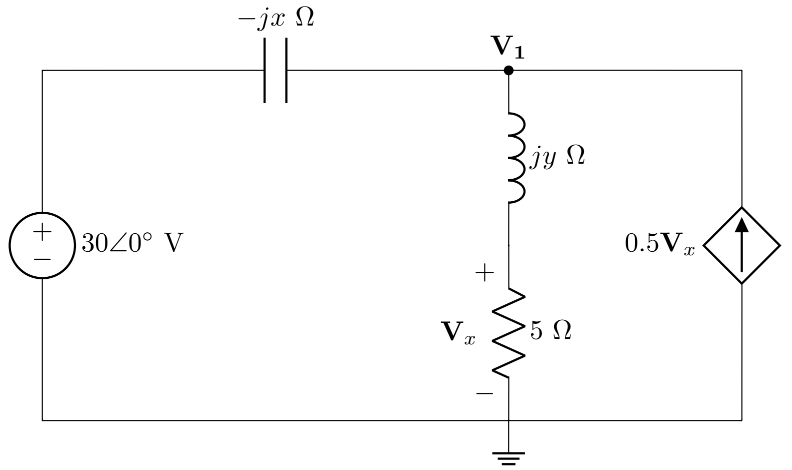
$= \dfrac{10^3 L}{j(0.1LC - 10^4)}$

$Z_2 = Z_R \| Z_1$ (use matlab)

$\Rightarrow V = j2 \cdot Z_2$ (use matlab).

39

Question 40 [2]	
Find $\mathbf{V}_1$ in the circuit below. The values x and y are given in ClickUP.	



40	(a) If $\mathbf{V}_1 = C + jD$, what is the real part C? [2]
40	(a) If $\mathbf{V}_1 = C + jD$, what is the imagainary part D? [2]

$$\frac{V_1 - 30}{-jX} + \frac{V_1}{5+jy} - 0.5 \cdot V_1 \cdot \frac{5}{5+jy} = 0$$

$$\frac{V_1 - 30}{-jX} = \frac{1.5V_1}{5+jy}$$

$$\Rightarrow -j1.5X\, V_1 = (5+jy)V_1 - 30(5+jy)$$

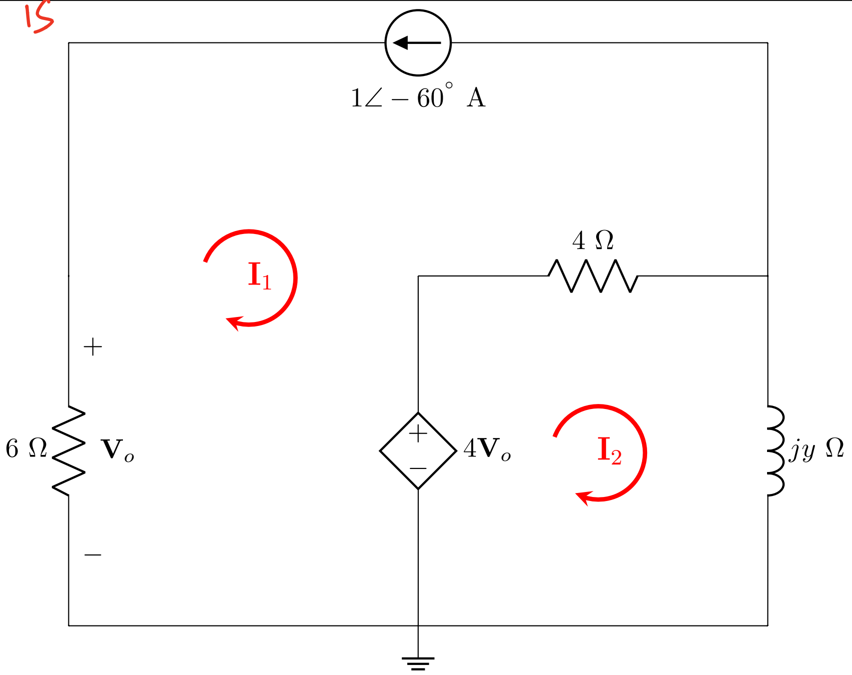
$$(5+jy-j1.5x)V_1 = 30(5+jy)$$

$$\Rightarrow V_1 = \frac{30(5+jy)}{5+j(y-1.5x)}$$

40

Question 41 [2]

In the circuit below, use **mesh** analysis to find $\mathbf{I_2}$.
The value~~s~~ of y ~~are~~ is given in ClickUP.



41	(a) If $\mathbf{I_2} = E + jF$, what is the real part E? [2]
41	(b) If $\mathbf{I_2} = E + jF$, what is the imaginary part F. [2]

$I_1 = -1\angle -60^\circ, \quad V_0 = 6\angle -60^\circ$

$-4V_0 + 4(I_2 - I_1) + jyI_2 = 0$

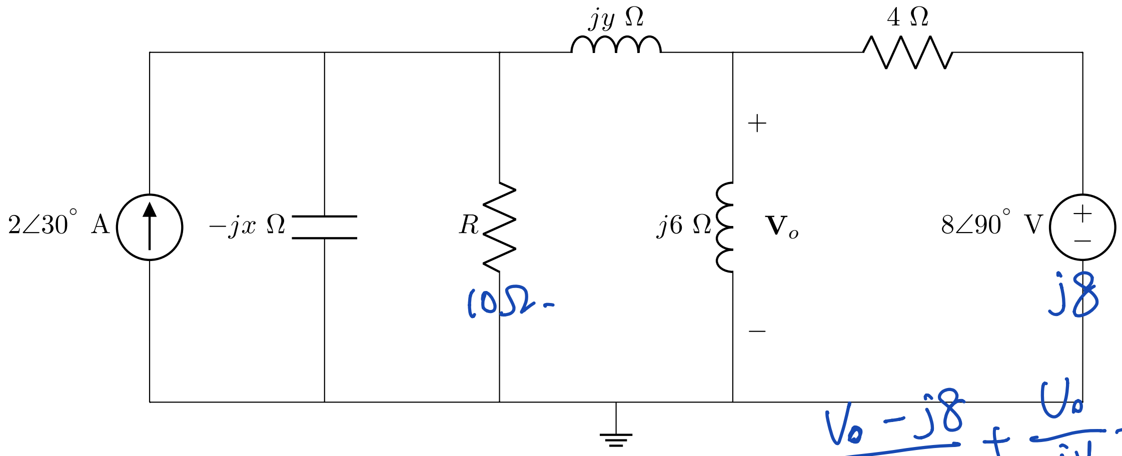
$\Rightarrow -24\angle -60^\circ + (4+jy)I_2 + 4\angle -60^\circ = 0$

$\Rightarrow (4+jy)I_2 = 20\angle -60^\circ$

41 $\quad I_2 = \dfrac{20\angle -60^\circ}{4+jy}$

Questions 42 to 44 [6]

In **Circuit A**, when $R = 0\ \Omega$, find the exclusive contribution, $\mathbf{V_o^1}$ from the voltage source to $\mathbf{V_o}$. The values x and y are given in ClickUP.



Circuit A

$$\frac{V_o - j8}{4} + \frac{V_o}{jy} + \frac{V_o}{j6} = 0$$

$$\left(\frac{1}{4} + \frac{1}{jy} + \frac{1}{j6}\right) V_o = j2$$

$$\Rightarrow V_o = \frac{j2}{\left(\frac{1}{4} + \frac{1}{jy} + \frac{1}{j6}\right)}$$

42

42	(a) If $\mathbf{V_o^1} = G + jH$, what is the real part G? [2]
42	(b) If $\mathbf{V_o^1} = G + jH$, what is the imaginary part H? [2]

When $R = 10\ \Omega$, the Circuit A is transformed to **Circuit B** using source transformation. Determine $\mathbf{I_1}$ and $\mathbf{Z_1}$.

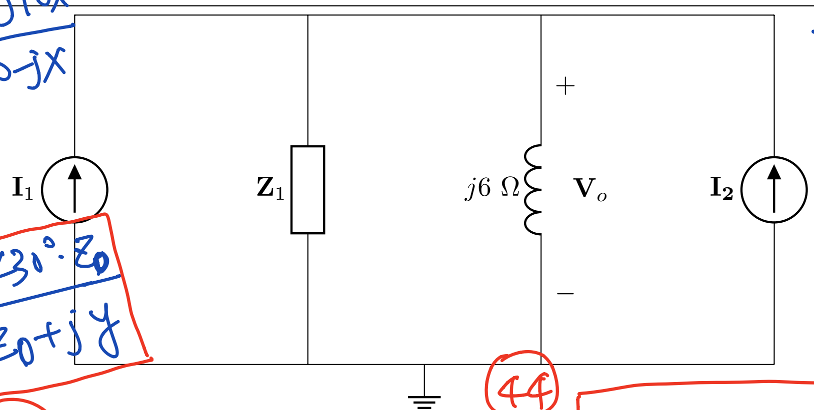
$$Z_0 = 10 \| -jx = \frac{-j10x}{10 - jx}$$

$$V_1 = 2\angle 30^\circ \cdot Z_0$$

$$Z_T = Z_0 + jy$$

$$\Rightarrow I_1 = \frac{V_1}{Z_T} = \boxed{\frac{2\angle 30^\circ \cdot Z_0}{Z_0 + jy}}$$

43



Circuit B

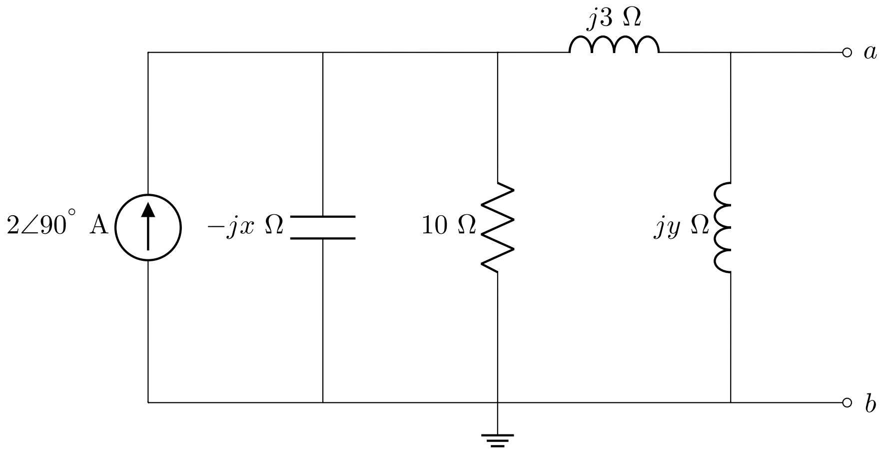
44

$$\boxed{Z_1 = Z_T \| 4}$$

43	(a) If $\mathbf{I_1} = K + jL$, what is the real part K? [2]
43	(b) If $\mathbf{I_1} = K + jL$, what is the imaginary part L? [2]
44	(a) If $\mathbf{Z_1} = P + jQ$, what is the real part P? [2]
44	(b) If $\mathbf{Z_1} = P + jQ$, what is the imaginary part Q? [2]

Questions 45 to 46 [4]

In the circuit below, find the Norton equivalent current $\mathbf{I_N}$ and impedance $\mathbf{Z_N}$ at terminals a-b. The values x and y are given in ClickUP.



45	(a) If $\mathbf{I_N} = S + jT$, what is the real part S? [2]
45	(b) If $\mathbf{I_N} = S + jT$, what is the imaginary part T? [2]
46	(a) If $\mathbf{Z_N} = U + jV$, what is the real part U? [2]
46	(b) If $\mathbf{Z_N} = U + jV$, what is the imaginary part V? [2]

45

$$Z_1 = -jx \,||\, 10 = \frac{-j10x}{10-jx}$$

$$Z_2 = Z_1 + j3, \quad Z_N = jy \,||\, Z_2.$$

46

$$I_N = j2 \cdot \frac{Z_1}{Z_1 + j3}.$$

Questions 47 to 48 [4]

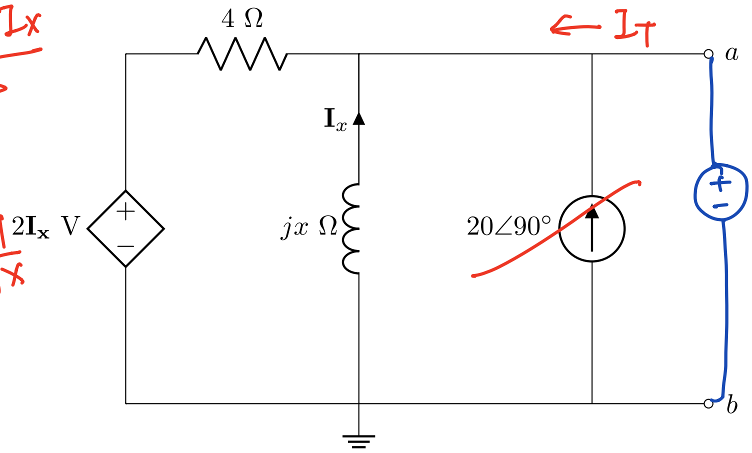
In the circuit below, find the Thévenin equivalent voltage $\mathbf{V_{Th}}$ and impedance $\mathbf{Z_{Th}}$ at terminals a-b. The value of x is given in ClickUP.

$I_T = -\frac{1}{jx} + \frac{1-2I_x}{4}$

$= -\frac{1}{jx} + \frac{1}{4} - \frac{I_x}{2}$

$= -\frac{1}{jx} + \frac{1}{4} + 0.5\frac{1}{jx}$

$= \frac{1}{4} - \frac{0.5}{jx}$



$I_x = -\frac{1}{jx}$

$1\angle 0^\circ$ V.

$\Rightarrow \boxed{Z_{Th} = \frac{1}{\frac{1}{4} - \frac{0.5}{jx}}}$

48

47	(a) If $\mathbf{V_{Th}} = X_1 + jY_1$, what is the real part X_1? [2]
47	(b) If $\mathbf{V_{Th}} = X_1 + jY_1$, what is the imaginary part Y_1? [2]
48	(a) If $\mathbf{Z_{Th}} = X_2 + jY_2$, what is the real part X_2? [2]
48	(b) If $\mathbf{Z_{Th}} = X_2 + jY_2$, what is the imaginary part Y_2? [2]

$$\frac{V_{Th} - 2I_x}{4} + \frac{V_{Th}}{jx} - j20 = 0$$

$$\Rightarrow \frac{V_{Th} - 2\cdot\left(-\frac{V_{Th}}{jx}\right)}{4} + \frac{V_{Th}}{jx} = j20$$

$$\frac{V_{Th}}{4} + 0.5\frac{V_{Th}}{jx} + \frac{V_{Th}}{jx} = j20$$

$$\left(\frac{1}{4} + \frac{1.5}{jx}\right) V_{Th} = j20 \Rightarrow V_{Th} = \frac{j20}{\frac{1}{4} + \frac{1.5}{jx}}$$

47

Total Section 3 / Totaal Afdeling 3: [32]

Grand Total: [96]